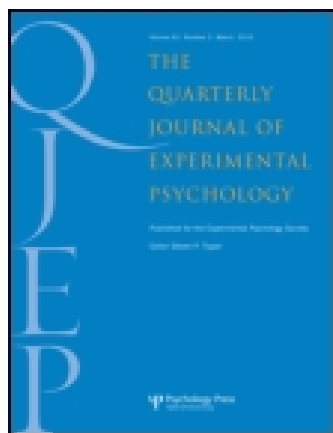


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### Northern=smart and Southern=nice: The development of accent attitudes in the United States

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# Northern = smart and Southern = nice: The development of accent attitudes in the United States

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Adults evaluate others based on their speech, yet little is known of the developmental trajectory by which accent attitudes are acquired. Here we investigate the development of American children's attitudes about Northern- and Southern-accented American English. Children in Illinois (the "North") and Tennessee (the "South") evaluated the social desirability, personality characteristics, and geographic origins of Northern- and Southern-accented individuals. Five- to 6-year-old children in Illinois preferred the Northern-accented speakers as potential friends, yet did not demonstrate knowledge of any stereotypes about the different groups; 5–6-year-old children in Tennessee did not show a preference towards either type of speaker. Nine- to 10-year-old children in both Illinois and Tennessee evaluated the Northern-accented individuals as sounding "smarter" and "in charge", and the Southern-accented individuals as sounding "nicer." Thus, older children endorse similar stereotypes to those observed in adulthood. These accent attitudes develop in parallel across children in different regions and reflect both positive and negative assessments of a child's own group.

**Keywords:** Social cognition; Language; Stereotypes; Accent attitudes; Social preferences.

Diverse studies of adults suggest that people categorize and make inferences about others based on their language and accent (Giles & Billings, 2004; Labov, 2006), and that speaking what is considered to be a dispreferred accent can be stigmatized (Gluszek & Dovidio, 2010; Lippi-Green, 1997). Accent attitudes in adults are typically thought to reflect knowledge of cultural stereotypes about different groups of individuals and resulting subjective judgements about their speech. The influence of cultural knowledge on accent attitudes

is particularly compelling in light of the predominant evidence of adults' awareness of linguistic stereotypes, as well as the paucity of experimental evidence demonstrating that (in the absence of a listener's cultural knowledge or familiarity with a language) some accents or dialects are intrinsically more aesthetically pleasing than others (Giles & Billings, 2004; Giles, Bourhis, & Davies, 1974; Giles, Bourhis, Trudgill, & Lewis, 1974; Giles, Harrison, Creber, Smith, & Freeman, 1983; Trudgill & Giles, 1983). Accent evaluations can

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take on myriad forms: Adults can profess preferences for the speech of ingroup members, preferences for speech that is considered high status, and also more complex evaluations of others' personalities, for instance in judging that someone is warm but not competent (see Giles & Billings, 2004, for a review).

Accent attitudes have garnered great attention from sociolinguists and psycholinguists (e.g., Cargile, Giles, Ryan, & Bradac, 1994; Giles & Billings, 2004; Labov, 2006). Nonetheless, studies of accent-based social evaluation are historically underrepresented by experimental psychologists who study intergroup attitudes and relations (Gluszek & Dovidio, 2010). Moreover, only a handful of studies have explored how language-based social evaluation develops across childhood (e.g., Anisfeld & Lambert, 1964; Day, 1980). These questions are ripe for study by experimental psychologists—including developmental psychologists—for both theoretical and practical reasons. First, research on the development of accent attitudes is of theoretical interest for our understanding of person perception. As illustration, one recent study provides intriguing evidence that adults' categorization of others based on their accent outweighs categorization based on visual cues to their ethnicity (Rakić, Steffens, & Mummendey, 2011). Initial research with young children suggests that children's social evaluations based on accent and language similarly have the potential to outweigh those based on race (Kinzler & Dautel, 2012; Kinzler, Dupoux, & Spelke, 2007; Kinzler, Shutts, DeJesus, & Spelke, 2009; Kinzler & Spelke, 2011). Research that explores the developmental trajectory of children's thinking about others based on their accent can inform a more general understanding of the development of social categorization.

Second, studies of the development of accent attitudes have real-world significance. Language-based stereotypes and social evaluation have potential consequences in many arenas of personal and professional interactions (Gluszek & Dovidio, 2010; Lippi-Green, 1997; Matsuda, 1991), and understanding the nature of accent attitudes could be beneficial in mitigating bias. Studies of the

development of accent attitudes will be particularly important for understanding when and under what circumstances these attitudes are transmitted and may be malleable. In the present research, we address this challenge through studies of the development of linguistic stereotypes in children living in Northern and Southern regions of the United States.

## NORTHERN VS. SOUTHERN AMERICAN ENGLISH

Though linguists acknowledge subtle gradations of accent variation in the United States, "folk" linguists often do not: When asked to draw a map of linguistic variation in the United States, American adults from diverse geographic origins draw remarkably similar, stratified maps of accent variation, with the "South" as the most consistently drawn area. In choosing states where people speak either "correctly" or "incorrectly", adults from the Northern United States (e.g., Michigan) reliably rank the South as an area in which people speak the "worst English" (Preston, 1993, 1998). Adults from the Southern United States (e.g., Mississippi) show a similar, though attenuated pattern of responses. Southerners rate their own region as relatively low in correctness, revealing what is termed "linguistic insecurity" (Labov, 2006; Preston, 1999), but they also rate speech from the South as high in pleasantness (Preston, 1998). Though these stereotypes have potential consequences for employment and self-evaluation in the US (Lippi-Green, 1997), the developmental trajectory by which accent attitudes are acquired is unclear, and it is unknown whether these attitudes emerge similarly or differentially among children living in the Northern and Southern areas of the United States.

Past research makes divergent predictions about how American children might reason about different dialects of American English. From one perspective, children in both the North and the South could express social preferences for individuals who speak in their local dialect. Research suggests that preferences for familiar languages emerge remarkably early in life (Mehler et al.,

1988). By five months of age, infants can discriminate among dialects of their native language (Nazzi, Jusczyk, & Johnson, 2000), and they look longer at individuals who previously spoke their language in a native, rather than a non-native, accent (Kinzler et al., 2007). Five-year-old children express social preferences for native-accented speakers over individuals who speak in a foreign accent (Kinzler et al., 2009; see also Kinzler & DeJesus, in press). Children also make inferences about people based on their language; for instance, they assume that individuals who speak a native language are more likely to be from a familiar racial group or to dress and live in familiar-style clothes and houses (Hirschfeld & Gelman, 1997; Wagner, Greene-Havas, & Gillespie, 2010). Given this past evidence, we might predict that—at least early in development—children across different environments would prefer whatever speech is native to their local community, and they might express these preferences across any domain of evaluation.

From another perspective, children in both locations might attend to linguistic stereotypes and status, and therefore prefer speech that they associate with the most favourable stereotypes, regardless of whether or not it is their local dialect (e.g., Lambert, Frankel, & Tucker, 1966; Lambert, Hodgson, Gardner, & Fillenbaum, 1960; Price, Fluck, & Giles, 1983). In support, past research finds that by 6 years of age, Hawaiian children from diverse linguistic backgrounds preferred “standard” English to Hawaiian Creole (Day, 1980). Furthermore, recent research in South Africa observed that 5–10-year old children who speak both Xhosa and English preferred English speakers to Xhosa speakers, even though Xhosa was their first language and the language spoken at home (Kinzler, Shutts, & Spelke, 2012). Nonetheless, past studies investigating children’s attention to linguistic status have been conducted primarily among children living in bilingual or bidialectal communities, where children have extensive experience interacting with individuals of different linguistic backgrounds. It is thus an open question whether linguistic stereotypes influence monolingual English-speaking American children’s evaluations of others.

A third possibility is that children may more readily endorse positive than negative stereotypes about their own group. If so, though children in both locations may gain knowledge of linguistic stereotypes, those stereotypes may not be the same or may not be acquired at the same point in development in different communities. Research with adults provides suggestions that this may be the case: Among adults, Northerners and Southerners both rate the positive features stereotypically associated with their speech community (e.g., “correct” or “pleasant”) as more extreme than the negative features (Preston, 1998, 1999), and high identification with a linguistic community is correlated with positive evaluation of speakers who are part of that community (Luhman, 1990). However, past research on accent attitudes in adults has rarely presented participants in different locations with identical stimuli or asked them to respond to identical questions. Thus, direct comparisons across different populations can be difficult.

Finally, it is possible that children may not detect differences in Northern vs. Southern speech at all. Past studies that evaluate American children’s social reasoning about language (e.g., Hirschfeld & Gelman, 1997; Kinzler et al., 2009; Wagner et al., 2010) have typically presented children with “foreign” languages and accents that were produced by non-native speakers of English (e.g., a native speaker of French, speaking either in French or in English with a French accent). It is possible that children’s early preferences for native-accented speakers might rely exclusively on a “native” vs. “foreign” distinction. Children may prefer a native speaker (e.g., American English) to a foreign speaker (e.g., French or French-accented English) yet may not express similar social preferences for a familiar vs. unfamiliar regional accent or dialect. Indeed, research conducted in Europe suggests that children have more difficulty detecting differences among regional accents than non-native accents (Flocchia, Butler, Girard, & Goslin, 2009; Girard, Flocchia, & Goslin, 2008).

The current research aims to explore the developmental trajectory of American children’s attitudes and evaluations of Northern vs. Southern American English speakers. In one study that

provides particular motivation for the current research, 5–6-year-old monolingual English-speaking American children from the Northeast expressed explicit social preferences for native-accented (Northern American English) over foreign-accented speakers (French-accented English; Kinzler et al., 2009). Using the same methods that previously revealed social preferences based on native vs. foreign accent among American monolingual English-speaking children, we presented children of two age groups (5–6- and 9–10-year-olds) in two locations (“Illinois, the “North,” and Tennessee, the “South”) with individuals paired with voice clips recorded in each location. Studies of children of two ages in two communities allowed us to explore two primary questions: (1) Do young children demonstrate social preferences for and expectations about individuals who speak in local vs. non-local regional accents? (2) Do accent stereotypes develop distinctly or in parallel among children living in the North and the South?

## EXPERIMENT 1

Experiment 1 presented 5–6-year-old children in both Illinois and Tennessee with images of individuals paired with Northern- or Southern-accented American English.<sup>1</sup> Participants’ friendship preferences, sociolinguistic evaluations, and expectations about individuals’ geographic origins were assessed. All children viewed the same stimuli; we therefore could test children’s responses in each location and draw comparisons across regions.

## Method

### *Participants*

Participants included two groups of children: 5–6-year-old monolingual English-speaking children from Chicago, Illinois ( $N = 24$ ; 12 girls, 12 boys; mean age = 71.7 months; range = 60.1–82.2 months; 62.5% White, 20.8% African American, 12.5% biracial, 4.2% Asian), and 5–6-year-old

monolingual English-speaking children from a small town in southern Tennessee, near the Alabama border ( $N = 24$ ; 12 girls, 12 boys; mean age = 74.5 months; range = 67.6–79.4 months; 95.8% White, 4.2% biracial).

Children tested in Illinois live in a dialect region that is described by phonologists as the “North” (more specifically, the “Inland North”). Additionally, perceptual dialectologists report that this area is consistently labelled by lay people as the “North” and as representative of the “best English” in America. In contrast, children tested in Tennessee live in a dialect region described by phonologists as the “South” (more specifically, the “Inland South”); this region is consistently labelled by lay people as the “South” and rated relatively low in “correctness” but relatively high in “pleasantness” (see Labov, Ash, & Boberg, 2006, for regional phonology maps; see Preston, 1998, for composite maps of adults’ ratings of dialect areas).

### *Materials*

Faces consisted of 16 racially ambiguous adults (eight female, eight male, each created by morphing a White face with a Black face) presented in gender-matched pairs against a white background on a laptop computer. Voice stimuli consisted of 16 voice clips of adults speaking in English and were recorded from speakers from Illinois or Tennessee using a high-quality voice recorder. Voice clips were approximately 3 s in length and neutral in content (e.g., “In general, dogs are bigger than cats”).

### *Procedure*

Children were first presented with eight Friendship trials to replicate the procedure of Kinzler et al. (2009). In each trial, the experimenter showed the child a pair of faces, pointed to each face, and played a voice clip of one Northern- and one Southern-accented speaker in turn. Children were asked, “Which one would you want to be friends with?” They next saw the same eight pairs of faces and voices a second time. On each of four trials in the Sociolinguistic evaluation block,

<sup>1</sup>We choose to refer to our language samples as “Northern” and “Southern”. It would also be appropriate to classify our Northern speech samples as “Standard American English”.

children were asked three questions (in counterbalanced order): “Who do you think is nicer?”, “Who do you think is smarter?”, and “Who do you think is in charge?”. On each of four trials in the Geography block, children were asked two questions (in counterbalanced order): “Who do you think lives around here?” and “Who do you think is American?”.

### Design

The order in which Northern- or Southern-accented voices were presented was counterbalanced within and across participants. Pairings of voice type (Northern- vs. Southern-accented) to a particular face, the order of the sociolinguistic evaluation and geography blocks, and the question order within each block were counterbalanced across participants. All children were tested by a local-accented experimenter.

## Results

### *Friendship block*

Five- and 6-year-old children in Illinois selected Northern-accented speakers as friends ( $M_{\text{north}} = 67.2\%$ ,  $SE = 3.75$ ),  $t(23) = 4.59$ ,  $p < .001$ ,  $d = 0.94$ , whereas the responses of 5–6-year-old children in Tennessee did not differ from chance ( $M_{\text{north}} = 53.1\%$ ,  $SE = 3.71$ ),  $t(23) = 0.84$ ,  $p = .41$ ,  $d = 0.17$ . The responses of the two groups differed significantly from one another, as children in Illinois were significantly more likely than children tested in Tennessee to select Northern-accented speakers as friends,  $F(1, 46) = 7.12$ ,  $p = .01$ ,  $\eta_p^2 = 0.13$ .

### *Sociolinguistic evaluation block*

Collapsing across the three sociolinguistic questions (“nicer,” “smarter,” and “in charge”), Illinoisan children demonstrated significantly more positive responses towards Northern speakers than towards Southern speakers ( $M_{\text{north}} = 57.6\%$ ,  $SE = 3.32$ ),  $t(23) = 2.30$ ,  $p = .031$ ,  $d = 0.47$ . Tennessean children’s responses again did not differ from chance ( $M_{\text{north}} = 50.7\%$ ,  $SE = 2.35$ ),  $t(23) = 0.30$ ,  $p = .77$ ,  $d = 0.06$ . A repeated-measures analysis of variance (ANOVA) with question type (nicer, smarter, in charge) as a

within-subjects factor and location (Illinois vs. Tennessee) as a between-subjects factor revealed a non-significant trend of location: Children in Illinois displayed slightly more positivity towards Northern-accented speakers than did children in Tennessee (57.6% vs. 50.7%),  $F(1, 46) = 2.91$ ,  $p = .095$ ,  $\eta_p^2 = 0.06$ , but there was no significant effect of question type,  $F(2, 92) = 0.48$ ,  $p = .62$ ,  $\eta_p^2 = 0.01$ , and no interaction between location and question type,  $F(2, 92) = 0.31$ ,  $p = .73$ ,  $\eta_p^2 = 0.007$ .

### *Geography block*

Collapsing across the two geography questions (“American” and “lives around here”), children in Illinois selected the Northern speakers, rather than the Southern speakers, as being local and American ( $M_{\text{north}} = 62.0\%$ ,  $SE = 4.96$ ),  $t(23) = 2.42$ ,  $p = .024$ ,  $d = 0.49$ . Tennessean children’s responses did not differ from chance ( $M_{\text{north}} = 50.5\%$ ,  $SE = 1.40$ ),  $t(23) = 0.37$ ,  $p = .71$ ,  $d = 0.076$ . A repeated-measures ANOVA with question type (lives around here vs. American) as a within-subjects factor and location (Illinois vs. Tennessee) as a between-subjects factor revealed a significant effect of location,  $F(1, 46) = 4.94$ ,  $p = .031$ ,  $\eta_p^2 = 0.097$ , but no effect of question type,  $F(1, 46) = 0.03$ ,  $p = .86$ ,  $\eta_p^2 = 0.001$ , and no interaction between location and question type,  $F(1, 46) = 0.00$ ,  $p = 1.00$ ,  $\eta_p^2 < 0.001$ .

Means and non-parametric results for children’s choices for each question type in each location are presented in Table 1 (top).

## Discussion

The results of Experiment 1 provide insight into children’s early social evaluations based on regional dialect. First, children tested in Illinois expressed positivity towards Northern-accented speech: They chose Northern individuals as friends, and also chose them as having more favourable personality attributes and as living in their geographical region. This result provides evidence that children’s social preference for native- over foreign-accented speakers (e.g., Kinzler et al.,

**Table 1.** Results from Experiments 1 and 2, divided by age group and location

|                                | % Northern choices | No. of participants<br>chose majority Northern | No. of participants<br>chose majority Southern | No. of participants<br>chose equally |
|--------------------------------|--------------------|------------------------------------------------|------------------------------------------------|--------------------------------------|
| <i>IL, 5–6-years (N = 24)</i>  |                    |                                                |                                                |                                      |
| Friendship                     | 67.2%              | 18                                             | 2                                              | 4                                    |
| Nicer                          | 56.3%              | 9                                              | 7                                              | 8                                    |
| Smarter                        | 55.2%              | 7                                              | 5                                              | 12                                   |
| In charge                      | 61.5%              | 10                                             | 5                                              | 9                                    |
| Lives here                     | 61.5%              | 13                                             | 5                                              | 6                                    |
| American                       | 62.5%              | 12                                             | 6                                              | 6                                    |
| <i>TN, 5–6-years (N = 24)</i>  |                    |                                                |                                                |                                      |
| Friendship                     | 53.1%              | 10                                             | 10                                             | 4                                    |
| Nicer                          | 46.9%              | 7                                              | 9                                              | 8                                    |
| Smarter                        | 53.1%              | 9                                              | 7                                              | 8                                    |
| In charge                      | 52.1%              | 8                                              | 6                                              | 10                                   |
| Lives here                     | 50.0%              | 8                                              | 6                                              | 10                                   |
| American                       | 51.0%              | 6                                              | 6                                              | 12                                   |
| <i>IL, 9–10-years (N = 24)</i> |                    |                                                |                                                |                                      |
| Friendship                     | 68.2%              | 15                                             | 4                                              | 5                                    |
| Nicer                          | 38.5%              | 4                                              | 11                                             | 9                                    |
| Smarter                        | 72.9%              | 16                                             | 2                                              | 6                                    |
| In charge                      | 58.3%              | 11                                             | 5                                              | 8                                    |
| Lives here                     | 82.3%              | 19                                             | 3                                              | 2                                    |
| American                       | 66.7%              | 15                                             | 6                                              | 3                                    |
| <i>TN, 9–10-years (N = 24)</i> |                    |                                                |                                                |                                      |
| Friendship                     | 56.3%              | 13                                             | 7                                              | 4                                    |
| Nicer                          | 38.4%              | 6                                              | 14                                             | 4                                    |
| Smarter                        | 65.6%              | 12                                             | 6                                              | 6                                    |
| In charge                      | 61.5%              | 11                                             | 5                                              | 8                                    |
| Lives here                     | 41.7%              | 10                                             | 12                                             | 2                                    |
| American                       | 57.3%              | 11                                             | 7                                              | 6                                    |

*Note:* Results are presented as the mean percentage of Northern speakers selected for each question type and also the number of children who selected a majority of Northern speakers (i.e., on more than half of trials per question type), children who selected a majority of Southern speakers, and children who selected equally.

2009) does not depend exclusively on a “native” vs. “foreign” distinction, but may also include a preference for a local vs. non-local regional dialect.

Children in Tennessee presented a different pattern of results. Overall, Tennessean children’s responses did not differ from chance. It is plausible that children in the South are exposed to both dialects and they therefore view both types of speech as equally favourable. For instance, children in tested in Tennessee may hear Northern-accented speech on a regular basis due to media exposure (Lippi-Green, 1997). Importantly, though, Illinoisan and

Tennessean children’s responses differed on measures of friendship, sociolinguistic evaluation, and geographical awareness, suggesting that relative experience with different dialects may influence young children’s social reasoning.

Interestingly, across both locations we observed no evidence that young children possess knowledge of the linguistic stereotypes that are commonly endorsed by adults about Northern vs. Southern American speech (i.e., Northerners as higher in status and intelligence, and Southerners as higher in warmth). Children’s responses to personality

questions that might be seen as laden with stereotypic content (e.g., nice vs. smart) did not vary systematically. This finding, in concert with the result that Northern children demonstrate a relative preference for speech with a more familiar accent, provides evidence that assessments of preferences can occur independently of knowledge of cultural stereotypes associated with a group of speakers.

## EXPERIMENT 2

Experiment 2 investigated potential changes in children's responses across development. We presented 9–10-year-old children in both Illinois and Tennessee with the same experimental paradigm as in the first experiment. Tests of two age groups of children across two communities allowed us to explore the development of accent attitudes, and whether they emerge in parallel or differentially across populations.

### Method

#### *Participants*

Participants included two groups of children: 9–10-year-old monolingual English-speaking children from Illinois ( $N=24$ ; 12 girls, 12 boys; mean age = 122.3 months; range = 109.5–127.6 months; 75.0% White, 20.8% African American, 4.2% Asian) and 9–10-year-old monolingual English-speaking children from Tennessee ( $N=24$ ; 12 girls, 12 boys; mean age = 121.5 months; range = 112.9–131.8 months; 83.3% White, 8.3% African American, 8.3% biracial or other). Participants were tested in the same dialect regions as in Experiment 1 (see Labov et al., 2006; Preston, 1998).

#### *Procedure*

The materials, procedure, and design were identical to Experiment 1.

### Results

#### *Friendship block*

Nine- and 10-year-olds in Chicago selected the Northern-accented speakers as friends ( $M_{\text{north}} =$

68.2%;  $SE = 4.82$ ),  $t(23) = 3.79$ ,  $p = .001$ ,  $d = 0.77$ . The responses of 9–10-year-old children in Tennessee did not significantly differ from chance ( $M_{\text{north}} = 56.2\%$ ;  $SE = 6.65$ ),  $t(23) = 0.94$ ,  $p = .36$ ,  $d = 0.19$ , and the two groups of children's responses did not differ significantly from each other,  $F(1, 46) = 2.13$ ,  $p = .15$ ,  $\eta_p^2 = 0.04$ . Comparing across ages, the responses of 9–10-year-old children in Illinois did not differ from 5–6-year-old children in Illinois,  $t(46) = -0.17$ ,  $p = .87$ ,  $d = 0.049$ ; the performance of older children in Tennessee also did not differ from that of younger children,  $t(46) = -0.41$ ,  $p = .68$ ,  $d = 0.12$ .

#### *Sociolinguistic evaluation block*

Collapsing across the three sociolinguistic questions ("nicer", "smarter," and "in charge"), children in Illinois chose the Northern-accented speakers more often than would be predicted by chance ( $M_{\text{north}} = 56.6\%$ ,  $SE = 2.79$ ),  $t(23) = 2.36$ ,  $p = .027$ ,  $d = 0.48$ . Tennessean children's responses demonstrated a similar trend ( $M_{\text{north}} = 55.2\%$ ,  $SE = 3.04$ ),  $t(23) = 1.71$ ,  $p = .10$ ,  $d = 0.35$ . More critically, a repeated-measures ANOVA with question type (nicer, smarter, in charge) as a within-subjects factor and location (Illinois vs. Tennessee) as a between-subjects factor revealed a robust effect of question type on children's judgements,  $F(2, 92) = 14.4$ ,  $p < .001$ ,  $\eta_p^2 = 0.24$ , yet no effect of location tested,  $F(1, 46) = 0.11$ ,  $p = .74$ ,  $\eta_p^2 = 0.002$ , and no interaction of question type with location,  $F(2, 92) = 0.42$ ,  $p = .66$ ,  $\eta_p^2 = 0.009$ . Across both locations, children selected the Northern-accented speakers as smarter ( $M_{\text{north}} = 69.3\%$ ,  $SE = 4.03$ ),  $t(48) = 4.79$ ,  $p < .001$ ,  $d = 0.69$ , and in charge ( $M_{\text{north}} = 59.9\%$ ,  $SE = 3.78$ ),  $t(48) = 2.62$ ,  $p = .012$ ,  $d = 0.38$ , yet selected the Southern-accented speakers as nicer ( $M_{\text{north}} = 38.5\%$ ,  $SE = 4.01$ ),  $t(48) = -2.86$ ,  $p = .006$ ,  $d = 0.41$ .

Comparing across ages, the effects of question type reported above interacted with age among children tested in Illinois,  $F(2, 92) = 5.51$ ,  $p = .006$ ,  $\eta_p^2 = 0.11$ . Older children tested in Illinois differentiated among sociolinguistic questions as described above (e.g.,  $M_{\text{north}} = 38.5\%$  for "nicer" and 72.9% for "smarter"), whereas younger Illinoisan children did not (e.g.,  $M_{\text{north}} = 56.3\%$

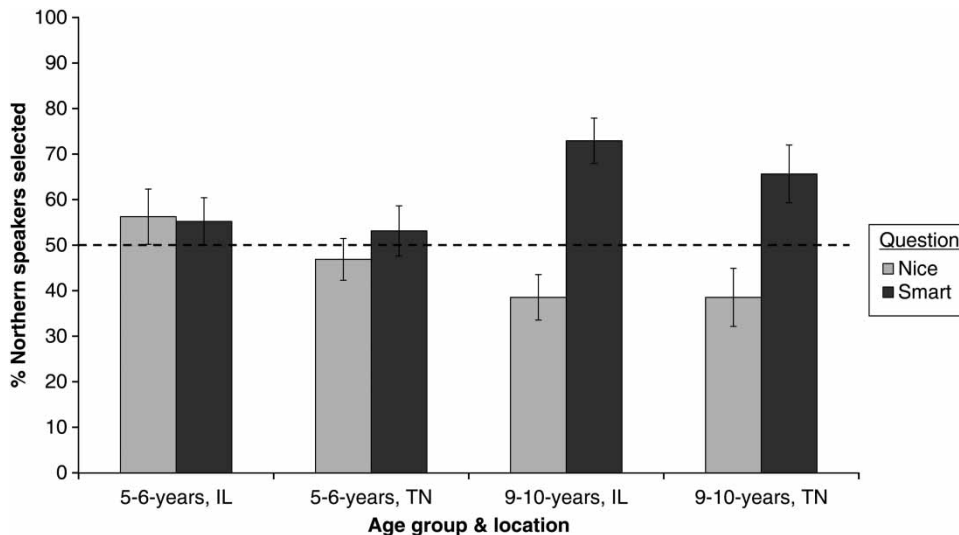


Figure 1. “Nice” vs. “smart” by age group and location.

for “nicer” and 55.2% for “smarter”). Although we see a similar descriptive pattern of responses when comparing older and younger children tested in Tennessee (e.g.,  $M_{\text{north}} = 38.4\%$  for “nicer” and 65.6% for “smarter” among older children;  $M_{\text{north}} = 46.9\%$  for “nicer” and 53.1% for “smarter” among younger children), we did not observe a significant interaction between age group and sociolinguistics question type for children tested in Tennessee,  $F(2, 92) = 1.87$ ,  $p = .16$ ,  $\eta_p^2 = 0.039$ . See Figure 1.

### Geography block

Collapsing across the two geography questions (“American” and “lives around here”), 9–10-year-old children in Illinois selected the Northern speakers, rather than the Southern speakers ( $M_{\text{north}} = 74.5\%$ ,  $SE = 4.35$ ),  $t(23) = 5.62$ ,  $p < .001$ ,  $d = 1.15$ , whereas 9–10-year-old Tennessean children’s responses did not differ from chance ( $M_{\text{north}} = 49.5\%$ ,  $SE = 5.85$ ),  $t(23) = -0.089$ ,  $p = .93$ ,  $d = 0.018$ . A repeated-measures ANOVA with question type (American vs. lives around here) as a within-subjects factor and location (Illinois vs. Tennessee) as a between-subject factor revealed a main effect of location (children in Illinois selected relatively more Northern voices,  $F(1, 23) = 11.8$ ,  $p = .001$ ,

$\eta_p^2 = 0.20$ , and also an interaction between location and question type,  $F(1, 46) = 6.28$ ,  $p = .016$ ,  $\eta_p^2 = 0.12$ . The difference between the two populations was greater for “who lives around here?” ( $M_{\text{north}} = 82.3\%$  in Illinois vs. 41.7% in Tennessee) than for “who is American?” ( $M_{\text{north}} = 66.7\%$  in Illinois vs. 57.3% in Tennessee).

Comparing across age groups, there was a marginally significant main effect of age group among children tested in Illinois,  $F(1, 46) = 3.59$ ,  $p = .065$ ,  $\eta_p^2 = 0.072$ , suggesting that older children were more slightly more likely than younger to select Northern speakers for geography-related questions ( $M_{\text{north}} = 74.5\%$  vs. 62.0%). No main effect of question type,  $F(1, 46) = 1.22$ ,  $p = .28$ ,  $\eta_p^2 = 0.026$ , or significant interaction between age group and geographic question type,  $F(1, 46) = 1.59$ ,  $p = .21$ ,  $\eta_p^2 = 0.033$ , was observed. Among children tested in Tennessee, there were no main effects of age group,  $F(1, 46) = 0.03$ ,  $p = .86$ ,  $\eta_p^2 = 0.001$ , or question type,  $F(1, 46) = 2.22$ ,  $p = .14$ ,  $\eta_p^2 = 0.046$ , and no significant interaction between age group and geographic question type,  $F(1, 46) = 1.70$ ,  $p = .199$ ,  $\eta_p^2 = 0.036$ .

See Table 1 for means and non-parametric results of children’s choices for each question type in each location.

## Discussion

The results of Experiment 2 reveal several interesting patterns of responses. First, although younger children's responses in Experiment 1 provided no evidence that they had knowledge of linguistic stereotypes (e.g., their assessments of who was "nice" and who was "smart" did not differ), older children tested in Experiment 2 differentiated among sociolinguistic questions. Nine- to 10-year-old children in both Illinois and Tennessee chose the Northerners as "smarter" and "in charge" and the Southerners as "nicer". This finding provides evidence that by 9 years of age, children endorse linguistic stereotypes that are similar to those observed in adulthood, and that these attitudes emerge in parallel among children living in different communities. Interestingly, we observed no effect of location on older children's sociolinguistic evaluations, and no interaction between location and question type. Thus, older children in both communities endorsed positive stereotypes about their own group, as well as stereotypes that deem their own group to be inferior. Regardless of their own group membership, children rated Northern-accented speakers as smarter and in charge and Southern-accented speakers as nicer.

The current results also reveal that in addition to expressing knowledge of linguistic stereotypes, 9- to 10-year-old children make other nuanced judgments about individuals based on their accent. The difference between the responses of Illinoisan and Tennessean children was greater when they were asked "who lives around here?" than when they were asked "who is American?". This interaction between location and geographic question type provides evidence that older children differentiate between knowledge of geographic proximity (i.e. "living around here") and nationality. It is also interesting to observe that older children's evaluations of who is "nice" and with whom they would "want to be friends" did not necessarily accord. Older children tested in Tennessee did not demonstrate a reliable friendship preference for speakers of either accent, yet they evaluated the Southern speakers as "nicer". Perhaps even

more notably, older children tested in Illinois rated Southern speakers as "nicer", yet they selected Northern speakers as potential friends. Open questions concern the structure of children's thinking about friendship across development, and to what extent it relies on familiarity, likeability, and status.

## GENERAL DISCUSSION

Testing children of two ages living in Illinois and Tennessee allowed us to observe the developmental trajectory by which accent attitudes are acquired among children living in different communities in the US. The data presented here provide evidence of two mechanisms that support children's linguistic evaluations of others. The first involves a preference for individuals who are perceived as being part of the child's native community—native, here, involves individuals who speak in a local versus a non-local regional accent. Five- to 6-year-old children in Illinois expressed social preferences for local, Northern-accented individuals, and their responses differed significantly from the responses of children in Tennessee (who did not express social preferences for either accent yet may be familiar with both). Neither of the two groups of young children, however, endorsed similar linguistic stereotypes to those endorsed by adults. Thus, early social preferences for native-accented speakers can exist in the absence of sophisticated knowledge of stereotypes pertaining to groups of speakers.

Accent attitudes towards Northern vs. Southern American English speech appear to emerge in the early school years, and to recruit burgeoning knowledge of cultural stereotypes. Nine- to 10-year-old children living in both Illinois and Tennessee reported that Northern-accented individuals sounded "smarter" and "in charge", whereas Southern-accented individuals sounded "nicer". Interestingly, we observed that these attitudes were acquired among children living in both Illinois and Tennessee. Consequently, children endorsed both positive and negative stereotypes about their own groups, instead of evaluating

their own group in a relatively more positive light for every attribute.

Though we document the development of accent attitudes in childhood here, critical unresolved questions concern how these attitudes are transmitted. Sociolinguistics research provides evidence that social perceptions of languages (e.g., “Italian is beautiful, and German is ugly”) are often the result of learned stereotypes, rather than inherent properties of the speech itself (Giles & Niedzielski, 1998). For instance, reported aesthetic differences among dialects of a language are not consistently acknowledged by naïve raters from a different linguistic or cultural group (Giles, Bourhis, & Davies 1974a, Giles, Bourhis, Trudgill, et al., 1974b; Trudgill & Giles, 1983). While the younger children we tested did not endorse assertions that Southern speech sounds nicer or that Northerner speech sounds smarter, by 9 years of age children across communities appear to have acquired that stereotype. How, then, are these attitudes communicated to children? As noted above, children in Tennessee may have exposure to both dialects, and that exposure may be skewed towards high-status Northern speakers in the form of national news anchors and film and television characters. In contrast, children in Illinois likely have little contact with Southern-accented speech. Given that research on the subtle transmission of social bias suggests that parents’ implicit racial attitudes can unintentionally be communicated to children (Castelli, Zogmaister, & Tomelleri, 2009), it seems possible that even a few casual comments about speakers’ accents (e.g., “he sounds friendly” or “he sounds like a snob”) might be sufficient to communicate bias to children. Moreover, children might even receive additional messages about Southern-accented speech from the media that is caricatured and reinforces the “Northern = smart, Southern = nice” stereotype (Lippi-Green, 1997).

Open questions concern the roles of familiarity and status in guiding children’s sociolinguistic evaluations. In Illinois, young children expressed social preferences for local-accented speakers in the absence of knowledge of cultural stereotypes about those groups. This finding of an early

“preference for local” accords with past research showing that young children prefer individuals who speak in their native language (Kinzler et al., 2007, Kinzler et al., 2009). Findings of early preferences for native speech have been observed even at ages at which infants do not yet speak themselves, suggesting that relative familiarity, rather than similarity to self, may guide children’s initial language-based social preferences. In Tennessee, young children did not differentiate between the two accents. As described above, one possible explanation for this finding is that they are exposed to both dialects, and thus both are treated as local. Nonetheless, it should be noted that children in Illinois were tested in an urban environment whereas Tennessee children were tested in a rural environment, where exposure to Northern-accented voices may stem primarily from media exposure rather than everyday interactions (Lippi-Green, 1997). In some cases, urban environments may afford greater exposure to diverse accents, which could in turn influence children’s earliest preferences for local speech. Research that explores the relative impact on children’s social preferences of different kinds of familiar speech—provided by family, peers, the community, and the media—would be of interest.

Considering status, it seems plausible that an early-to-develop preference for local speech does not depend on notions of prestige or status. Nevertheless, initial evidence suggests that reasoning about at least some kinds of status relationships (in particular those based on cues to size or physical dominance) can emerge as early as infancy (Mascaro & Csibra, 2012; Thomsen, Frankenhuys, Ingold-Smith, & Carey, 2011). By later in childhood, children infer status information based on subtle cues such as patterns of eye gaze among individuals (Chudek, Heller, Birch, & Henrich, 2012). Evidence also suggests that children also begin to associate racial groups with social status and wealth (e.g., Bigler, Averhart, & Liben, 2003; Olson, Shutts, Kinzler, & Weisman, in press). As illustration, research on the development of race-based social attitudes in childhood provides evidence of an asymmetry in social preferences, whereby majority-race children express greater ingroup

preferences than do minority-race individuals (Aboud, 1988; Aboud & Skerry, 1984; Davey, 1983; Shutts, Kinzler, Katz, Tredoux, & Spelke, 2011). The research presented here provides evidence that children begin to see accent as indicative of who is “in charge” at least by age nine. Open questions concern how children’s reasoning expressed here may relate to their thinking about status along other dimensions (e.g., wealth or education), how and whether children’s thinking about accent and race interact, and whether children’s thinking about correlations between patterns of speech and social class or status may emerge earlier in some environments.

It is also important to consider that children’s linguistic attitudes may have consequential (and potentially negative) impacts on their self-perception and academic success. Evidence suggests that adults feel insecure about speaking what they consider to be a non-standard accent (Labov, 2006; Preston, 1999). It is possible that children experience similar insecurities about their speech, though this question is as yet largely unexplored. Research on gender stereotypes suggests that women and girls are susceptible to stereotype threat when engaged in mathematics—namely, they underperform in situations where their gender is made salient (Ambady, Shih, Kim, & Pittinsky, 2001; Shih, Pittinsky, & Ambady, 1999). Moreover, teachers’ insecurities about their performance in math can be inadvertently communicated to students, with potentially adverse consequences for academic success (Beilock, Gunderson, Ramirez, & Levine, 2010). Do children who view their accent as sounding “less smart” experience similar insecurities about their potential as scholars? Might parents’ or teachers’ insecurities about their own accents be unwittingly communicated to children?

More generally, from an anthropological perspective, accent is hypothesized to provide a valid and reliable marker of coalitional group membership, which in turn fosters collaboration and affiliation towards ingroup individuals (Cohen, in press; Henrich & Henrich, 2007). Past research provides evidence that children selectively imitate the actions of native-accented individuals (Kinzler, Corriveau,

& Harris, 2011). Preschool-aged children interpret word labels as conventional when labels were provided by native speakers but not by foreign speakers (Koenig & Woodward, 2012; see also Behrend, 2009), and they preferentially learn the names of objects that they think of as being from their society versus from a foreign place (Henderson, Sabbagh, & Woodward, in press). Future research might investigate how and whether regional accent or dialect fosters collaboration, teaching, and learning among ingroup speakers.

Finally, social psychology has focused on race, gender, and age as primary social categories that guide individuals’ person perception (e.g., Fiske, 1998). Fortunately, discrimination based on these categories is prohibited by law in the United States. Protection against accent discrimination, however, does not benefit from similar legal scrutiny (Matsuda, 1991). The finding that even school-aged children in both the Northern and Southern US endorse linguistic stereotypes and think of Southern speech as being “less smart” suggests that accent-based social bias is early-forming and consequential. We hope that the research presented here might contribute to efforts to construct policy that acknowledges the potency of accent-based social evaluation and its potential implications for social bias and discrimination.

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## REFERENCES

- Aboud, F. (1988). *Children and prejudice*. Cambridge, MA: Basil Blackwell.
- Aboud, F. E., & Skerry, S. A. (1984). The development of ethnic attitudes: A critical review. *Journal of Cross-Cultural Psychology*, 15, 3–34.
- Ambady, N., Shih, M., Kim, A., & Pittinsky, T. L. (2001). Stereotype susceptibility in children: Effects of identity activation on quantitative performance. *Psychological Science*, 12, 385–390.

- Anisfeld, E., & Lambert, W. E. (1964). Evaluational reactions of bilingual and monolingual children to spoken languages. *Journal of Abnormal and Social Psychology*, 69, 89–97.
- Behrend, D. A. (2009, April). *Children's trust of foreign language speakers during word learning*. Talk presented as part of symposium on Children's Willingness to Accept Object Labels from Native and Foreign Speakers. The Biennial Meeting of the Society for Research in Child Development, Denver, CO.
- Beilock, S. L., Gunderson, E. A., Ramirez, G., & Levine, S. C. (2010). Female teachers' math anxiety affects girls' math achievement. *Proceedings of the National Academy of Sciences of the United States of America*, 107, 1860–1863.
- Bigler, R. S., Averhart, C. J., & Liben, L. S. (2003). Race and the workforce: Occupational status, aspirations, and stereotyping among African American children. *Developmental Psychology*, 39, 572–580.
- Cargile, A. C., Giles, H., Ryan, E. B., & Bradac, J. J. (1994). Language attitudes as a social process: A conceptual model and new directions. *Language and Communication*, 14, 211–236.
- Castelli, L., Zogmaister, C., & Tomelleri, S. (2009). The transmission of racial attitudes within the family. *Developmental Psychology*, 45, 586–591.
- Chudek, M., Heller, S., Birch, S., & Henrich, J. (2012). Prestige-biased cultural learning: Bystander's differential attention to potential models influences children's learning. *Evolution and Human Behavior*, 33, 46–56.
- Cohen, E. (in press). The evolution of tag-based cooperation in humans: The case for accent. *Current Anthropology*.
- Davey, A. (1983). *Learning to be prejudiced*. London, UK: Edward Arnold.
- Day, R. R. (1980). The development of linguistic attitudes and preferences. *TESOL Quarterly*, 14, 27–37.
- Fiske, S. T. (1998). Stereotyping, prejudice, and discrimination. In D. T. Gilbert, S. T. Fiske, & L. Gardner (Eds.), *The handbook of social psychology* (pp. 357–411). New York, NY: McGraw-Hill.
- Floccia, C., Butler, J., Girard, F., & Goslin, J. (2009). Categorization of regional and foreign accents in 5- to 7-year-old British children. *International Journal of Behavioral Development*, 33, 336–375.
- Giles, H., & Billings, A. (2004). Language attitudes. In A. Davies & E. Elder (Eds.), *The handbook of applied linguistics* (pp. 187–209). Oxford, UK: Blackwell.
- Giles, H., Bourhis, R., & Davies, A. (1974). Prestige speech styles: The imposed norm and inherent value hypothesis. In W. McCormack & S. Wurm (Eds.), *Language in anthropology. IV: Language in many ways*. The Hague, The Netherlands: Mouton.
- Giles, H., Bourhis, R., Trudgill, P., & Lewis, A. (1974). The imposed norm hypothesis: A validation. *Quarterly Journal of Speech*, 60, 405–410.
- Giles, H., Harrison, C., Creber, C., Smith, P. M., & Freeman, N. H. (1983). Developmental and contextual aspects of children's language attitudes. *Language & Communication*, 3, 141–146.
- Giles, H., & Niedzielski, N. (1998). Italian is beautiful, German is ugly. In L. Bauer & P. Trudgill (Eds.), *Language myths* (pp. 85–93). London, UK: Penguin Books.
- Girard, F., Floccia, C., & Goslin, J. (2008). Perception and awareness of accents in young children. *British Journal of Developmental Psychology*, 26, 409–433.
- Gluszek, A., & Dovidio, J. F. (2010). The way they speak: A social psychological perspective on the stigma of nonnative accents in communication. *Personality and Social Psychology Review*, 14, 214–237.
- Henderson, A. M. E., Sabbagh, M., & Woodward, A. L. (in press). Preschoolers' selective learning is guided by the principle of relevance. *Cognition*.
- Henrich, N., & Henrich, J. (2007). *Why humans cooperate: A cultural and evolutionary explanation*. Oxford, UK: Oxford University Press.
- Hirschfeld, L. A., & Gelman, S. A. (1997). What young children think about the relationship between language variation and social difference. *Cognitive Development*, 12, 213–238.
- Kinzler, K. D., Corriveau, K. H., & Harris, P. L. (2011). Children's selective trust in native-accent speakers. *Developmental Science*, 14, 106–111.
- Kinzler, K. D., & Dautel, J. B. (2012). Children's essentialist reasoning about language and race. *Developmental Science*, 15, 131–138.
- Kinzler, K. D., & DeJesus, J. M. (in press). Children's sociolinguistic evaluations of nice foreigners and mean Americans. *Developmental Psychology*. Advance online publication. doi: 10.1037/a0028740
- Kinzler, K. D., Dupoux, E., & Spelke, E. S. (2007). The native language of social cognition. *Proceedings of the National Academy of Sciences of the United States of America*, 104, 12577–12580.
- Kinzler, K. D., Shutts, K., DeJesus, J., & Spelke, E. S. (2009). Accent trumps race in guiding children's social preferences. *Social Cognition*, 27, 623–634.

- Kinzler, K. D., Shutts, K., & Spelke, E. S. (2012). Language-based social preferences among multilingual children in South Africa. *Language Learning and Development*, 8, 215–232.
- Kinzler, K. D., & Spelke, E. S. (2011). Do infants show social preferences for people differing in race? *Cognition*, 119, 1–9.
- Koenig, M., & Woodward, A. L. (2012). Toddlers learn words in a foreign language: The role of native vocabulary knowledge. *Journal of Child Language*, 39, 322–337.
- Labov, W. (2006). *The social stratification of English in New York City* (2nd ed.). New York, NY: Cambridge University Press.
- Labov, W., Ash, S., & Boberg, C. (2006). *The atlas of North American English: Phonetics, phonology, and sound change*. Berlin, Germany: Mouton de Gruyter.
- Lambert, W. E., Frankel, H., & Tucker, G. R. (1966). Judging personality through speech: A French-Canadian example. *Journal of Communication*, 16, 305–321.
- Lambert, W. E., Hodgson, R., Gardner, R., & Fillenbaum, S. (1960). Evaluational reactions to spoken languages. *Journal of Abnormal and Social Psychology*, 60, 44–51.
- Lippi-Green, R. (1997). *English with an accent: Language, ideology, and discrimination in the United States*. New York, NY: Routledge.
- Luhman, R. (1990). Appalachian English stereotypes: Language attitudes in Kentucky. *Language in Society*, 19, 331–348.
- Mascaro, O., & Csibra, G. (2012). Representation of stable social dominance relations by human infants. *Proceedings of the National Academy of Sciences of the United States of America*, 109, 6862–6867.
- Matsuda, M. J. (1991). Voices of America: Accent, anti-discrimination law, and a jurisprudence for the last reconstruction. *Yale Law Journal*, 100, 1329–1467.
- Mehler, J., Jusczyk, P., Lambertz, G., Halsted, N., Bertoni, J., & Amiel-Tison, C. (1988). A precursor of language acquisition in young infants. *Cognition*, 29, 144–178.
- Nazzi, T., Jusczyk, P., & Johnson, E. (2000). Language discrimination by English-learning 5-month-olds: Effects of rhythm and familiarity. *Journal of Memory and Language*, 43, 1–19.
- Olson, K., Shutts, K., Kinzler, K. D., & Weisman, K. (in press). Children associate racial groups with wealth: Evidence from South Africa. *Child Development*. Advance online publication. doi: 10.1111/j.1467-8624.2012.01819.x
- Preston, D. R. (1993). Folk dialectology. In D. Preston (Ed.), *American dialect research* (pp. 333–378). Philadelphia, PA: John Benjamins.
- Preston, D. R. (1998). They speak really bad English down south and in New York City. In L. Bauer & P. Trudgill (Eds.), *Language myths* (pp. 139–149). London, UK: Penguin Books.
- Preston, D. R. (1999). A language attitude approach to the perception of regional variety. In D. R. Preston (Ed.), *Handbook of perceptual dialectology* (pp. 359–373). Amsterdam, The Netherlands: Benjamins.
- Price, S., Fluck, M., & Giles, H. (1983). The effects of language of testing on bilingual pre-adolescents' attitudes towards Welsh and varieties of English. *Journal of Multilingual and Multicultural Development*, 4(2–3), 149–161.
- Rakić, T., Steffens, M. C., & Mummendey, A. (2011). Blinded by the accent! The minor role of looks in ethnic categorization. *Journal of Personality and Social Psychology*, 100, 16–29.
- Shih, M., Pittinsky, T. L., & Ambady, N. (1999). Stereotype susceptibility: Identity salience and shifts in quantitative performance. *Psychological Science*, 10, 80–83.
- Shutts, K., Kinzler, K. D., Katz, R. C., Tredoux, C., & Spelke, E. S. (2011). Race preferences in children: Insights from South Africa. *Developmental Science*, 14, 1283–1291.
- Thomsen, L., Frankenhuys, W. E., Ingold-Smith, M., & Carey, S. (2011). Big and mighty: Preverbal infants mentally represent social dominance. *Science*, 331, 477–480.
- Trudgill, P., & Giles, H. (1983). Sociolinguistics and linguistic value judgements. In P. Trudgill (Ed.), *On dialect: Social and geographical perspectives* (pp. 201–225). New York: New York University Press.
- Wagner, L., Greene-Havas, M., & Gillespie, R. (2010). Development in children's comprehension of linguistic register. *Child Development*, 81(6), 1678–1686.